

Capstone Project Report

Problem Statement

Many small colleges still evaluate student capstone reports entirely by hand, which is slow and inconsistent across faculty reviewers. This project proposes an AI-assisted pre-review tool that checks a report's literature survey, architecture diagram, and module alignment before a human reviewer sees it, so faculty time is spent on judgment rather than bookkeeping.

Literature Survey

[1] A. Kumar, "Automated Essay Scoring Using Transformer Models," IEEE Access, 2022.

[2] R. Chen and T. Osei, "A Survey of LLM-Based Document Review Systems," ACM Computing Surveys, 2023.

Prior work in automated review has focused mostly on essay scoring rather than multi-part technical reports, leaving a gap this project addresses.

System Modules

The system consists of a Document Parsing module that extracts sections and diagrams from the uploaded PDF, a Literature Survey module that checks citation relevance and recency, an Architecture Structure module that reads the submitted diagram, an Alignment module that compares the diagram against the module text, and a Report Aggregator module that produces the final composite score and downloadable report.

System Architecture

A block diagram below shows the flow between modules.

Parser

Scoring Engine

Report Output